

PROFESSIONAL PROFILE

PhD candidate and applied scientist with 7+ years of experience in data science and software development, specializing in conversational search and recommender systems. Expertise in natural language processing (NLP), large language models (LLMs), retrieval-augmented generation (RAG), and deep learning (DL). Proven record of publishing at top information retrieval conferences (SIGIR, RecSys, WWW, WSDM) and designing, training, and productionizing ML models at scale.

TECHNICAL SKILLS

- **Programming:** Python (Pandas, PySpark, scikit-learn, PyTorch), FastAPI, SQL, C#, JavaScript/TypeScript, React
- **ML & GenAI:** NLP, LLM fine-tuning, RAG, embeddings, retrieval (dense/sparse), recommender systems, A/B testing
- **Data & Search Systems:** Spark, Elasticsearch, MongoDB, ETL/ELT pipelines
- **Cloud & MLOps:** AWS (SageMaker, Bedrock, EC2, EMR, ECS, Lambda), Azure, CI/CD (GitHub Actions), W&B
- **Visualization & Tools:** Power BI, Jupyter, Docker, Git, LaTeX
- **Languages:** Croatian (native), English (fluent), Norwegian (professional), Spanish (basic)

EDUCATION

- **University of Stavanger** Stavanger, Norway
PhD Fellowship in Conversational AI Jul 2021 – Present
Supervisors: Prof. Krisztian Balog (UiS, Google), Dr. Filip Radlinski (Google)
- **University of Stavanger** Stavanger, Norway
Master of Science, Applied Data Science 2019 – 2021
- **University of Stavanger** Stavanger, Norway
Bachelor of Engineering, Control Engineering (Automation and Electronics Design) 2018 – 2019
- **University of Stavanger** Stavanger, Norway
Norwegian language and culture for international students 2017 – 2018

INDUSTRY EXPERIENCE

- **Amazon** Madrid, Spain
Applied Scientist Intern Aug 2024 – Feb 2025
 - **Book Recommendation Algorithms:** Designed and evaluated recommendation strategies leveraging content-based signals and LLMs for millions of customers using Spark and PySpark.
 - **Evaluation:** Designed new offline metrics for multi-widget recommendation that became the broader team’s core evaluation signal.
 - **ELT on Spark SQL:** Built ELT pipelines on AWS using Spark SQL on EMR to transform large-scale behavioral and content data into datasets for training and offline evaluation.
 - **Productionizing & Scaling:** Led productionization and A/B readiness. Owned data prep via ELT on Spark SQL (EMR) and large-scale experimentation pipelines on AWS (SageMaker, EMR, ECS, Lambda).
- **Presight Solutions** Stavanger, Norway
Data Scientist (part-time) Jun 2020 – Aug 2024
 - **RAG for Document Search:** Prototyped a Retrieval Augmented Generation system in Python, using Azure-hosted LLM endpoints for inference and MongoDB as the document store and retrieval backend.
 - **Predictive Models for Offshore Safety:** Built and validated ML models for offshore safety using heterogeneous data sources, including text, categorical, time-series, and sensor data.
 - **Product Search:** Implemented and maintained an information retrieval system for product search and discovery.
- **Presight Solutions** Stavanger, Norway
Software Engineer (part-time) Sep 2018 – Jun 2020
 - **Full-cycle Software Development:** Built dashboards, analytics, and data visualizations; wrote SQL queries and backend logic; developed and maintained full-stack features using .NET, React, and TypeScript.
- **Presight Solutions** Stavanger, Norway
R&D Intern Summer 2018
 - **Smart City Demo:** Designed a miniature smart city featuring embedded systems for sensors and wireless communication; showcased at Nordic Edge Expo 2018.

RESEARCH EXPERIENCE

- **University of Stavanger** Stavanger, Norway
Estimating User Domain Knowledge in Conversational Recommenders [arXiv] [GitHub] Fall 2025
 - **RAG-based Conversational Recommender System:** Developed a LangChain- and Streamlit-based conversational recommender system for running user studies and collecting interaction data in knowledge-driven RAG settings.
- **University of Stavanger** Stavanger, Norway
LiveRAG Challenge (SIGIR 2025) [arXiv][GitHub] Spring 2025
 - **Real-time RAG QA:** Participated in a team competition to build a real-time RAG system; Improved recall by 25% with advanced query rewriting methods and reduced latency through queuing and multithreaded execution.
- **University of Stavanger** Stavanger, Norway
Conversational Styles for Preference Elicitation (UMAP 2025) [doi] [GitHub] Spring 2024
 - **Conversational User Study:** Built a conversational recommender and ran a study in the scholarly domain, investigating how conversational style affects elicitation quality, task completion, and satisfaction.
- **University of Stavanger** Stavanger, Norway
PKG API: Personal Knowledge Graph Management (WWW 2024) [doi][GitHub] Spring 2024
 - **Knowledge Graph API Development:** Contributed to the design and implementation of an API for managing personal knowledge graphs, enabling personalised retrieval and recommendation scenarios.
- **University of Stavanger** Stavanger, Norway
Multi-query Rewriting for Conversational Passage Retrieval (SIGIR 2024) [doi] [GitHub] Fall 2023
 - **State-of-the-art Retrieval:** Proposed an effective extension to state-of-the-art query rewriting strategies that improves recall performance of up to 26% for conversational passage retrieval.
- **University of Stavanger** Stavanger, Norway
IAI MovieBot 2.0 (WSDM 2024) [doi] [GitHub] Spring 2023
 - **Neural Components:** Fine-tuned a sequence-to-sequence model for user intent understanding and trained a neural network with reinforcement learning to optimize the dialogue policy for personalized recommendation.
 - **Conversational UI Client:** Developed a React, embed-ready chat widget for interactive conversational interfaces. [GitHub]
- **University of Stavanger** Stavanger, Norway
DAGFiNN: A Conversational Conference Assistant (RecSys 2022) [doi] [GitHub] Spring 2022
 - **Multi-modal Conversational System:** Supervised a team of Master's and Bachelor's students to build a multi-domain conversational assistant on the Furhat social robot platform, integrating speech, text, and visual interfaces. Showcased at ECIR 2022 and published at RecSys 2022.
- **University of Stavanger** Stavanger, Norway
Preference Elicitation via Usage-related Questions (RecSys 2021, TORS 2023) [doi] [GitHub] Spring 2021
 - **Preference Elicitation Modeling:** Trained a transformer-based model to generate usage-related questions for conversational recommenders, supported by a rigorous multi-stage annotation protocol.

TEACHING

- **University of Stavanger** Stavanger, Norway
Lecturer, DAT640: Information Retrieval and Text Mining Fall 2022
 - **Course Owner:** Created and delivered lectures, designed assignments and projects, and authored and graded the final exam. Topics included search engine architecture, text pre-processing and indexing, retrieval models and evaluation, web search, semantic search, text clustering and categorization.
- **University of Stavanger** Stavanger, Norway
Teaching Assistant, DAT640: Information Retrieval and Text Mining Fall 2021
 - **Assignment Development:** Created graded assignments and group projects, guided students through solutions, and evaluated submissions. Developed an automated evaluation framework using Pytest.
- **University of Stavanger** Stavanger, Norway
Teaching Assistant, ELE520: Machine Learning Spring 2019 – Spring 2024
 - **Theory and Practice:** Guided students through theoretical and practical assignments, and evaluated submissions. Topics included Bayes decision theory, parametric and non-parametric estimation, linear discriminant functions, gradient descent, neural networks, and clustering.

SERVICE AND ACTIVITIES

- Reviewer: SIGIR, ECIR, WWW, WSDM, Computer Speech & Language Journal
- Student volunteer: ICTIR 2020, SIGIR 2021, ECIR 2022, SIGIR 2024
- Summer School on Recommender Systems, Copenhagen 2023